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Pharmaceutical Analysis Group Assignment

Principles and procedures in different types of redox titration

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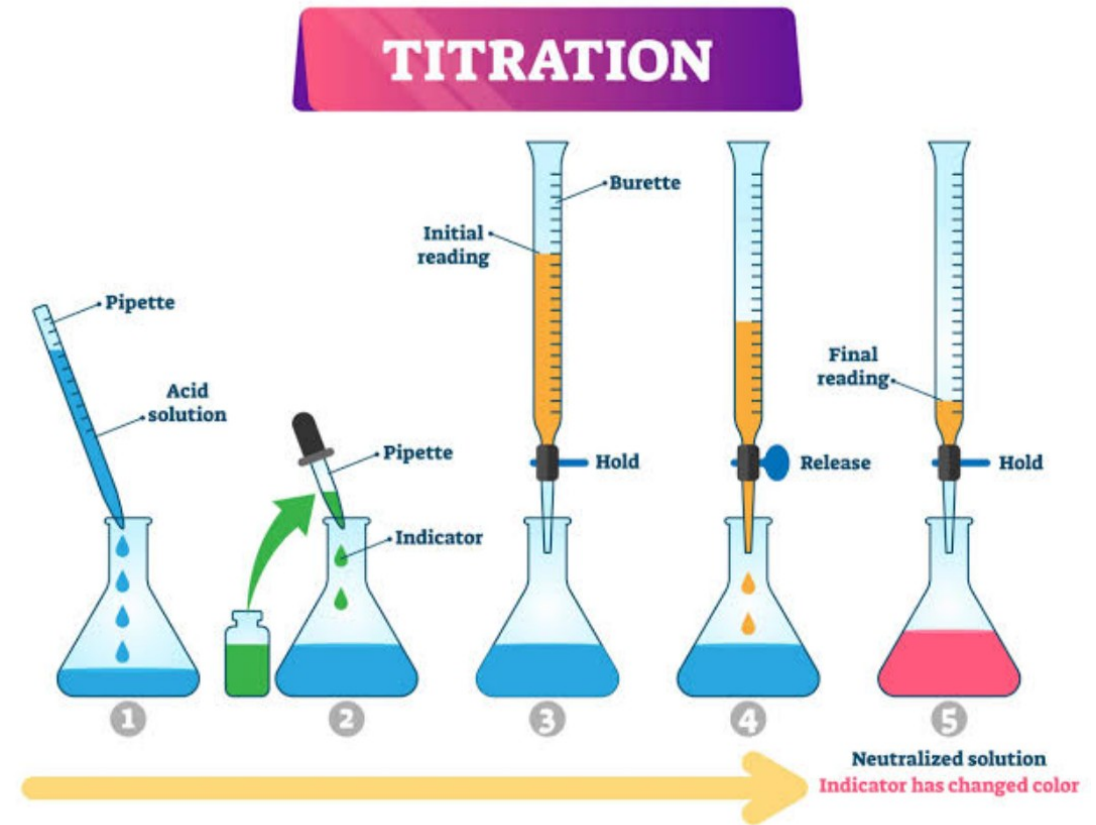
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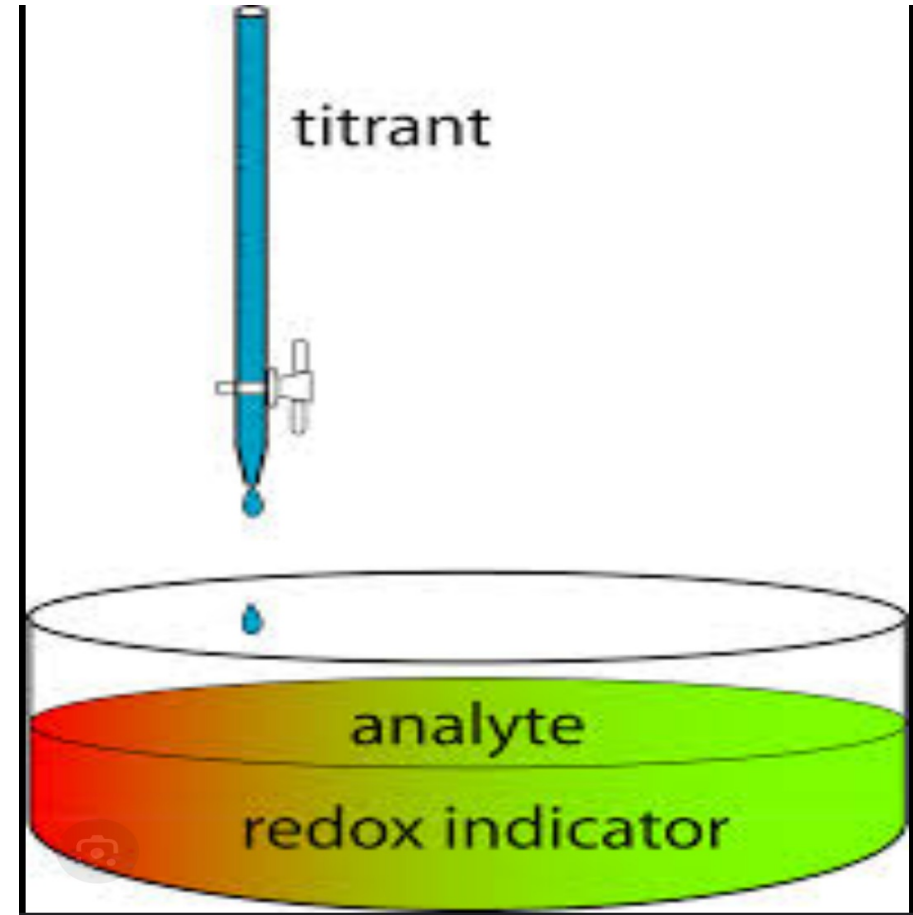
TITRATION

- Titration is an analytical technique used to determine the concentration of an unknown solution (the analyte) by reacting it with a solution of known concentration (the titrant).



REDOX TITRATION

- Redox titration is an analytical technique used to determine the concentration of an unknown solution (the analyte) by reacting it with a solution of known concentration (the titrant) through a redox reaction, which involves the transfer of electrons between the species being titrated.



Key Components:

- **Redox Reaction:** Involves oxidation (loss of electrons) and reduction (gain of electrons) processes.
- **Titrant:** A reagent with a known concentration that can either oxidize or reduce the analyte.
- **Analyte:** is the chemical that undergoes oxidation or reduction during the reaction
- **Indicator:** A substance that changes color at the end point or an electrode to detect the end point.
- **Equivalence Point:** The point at which the amount of titrant added is stoichiometrically equivalent to the amount of analyte present in the solution.

REDOX INDICATORS

- **Redox indicators** are substances that change color at a specific electrode potential, indicating the end point of a redox titration. They are crucial for visually signaling when the titration is complete.

Common redox indicators include:

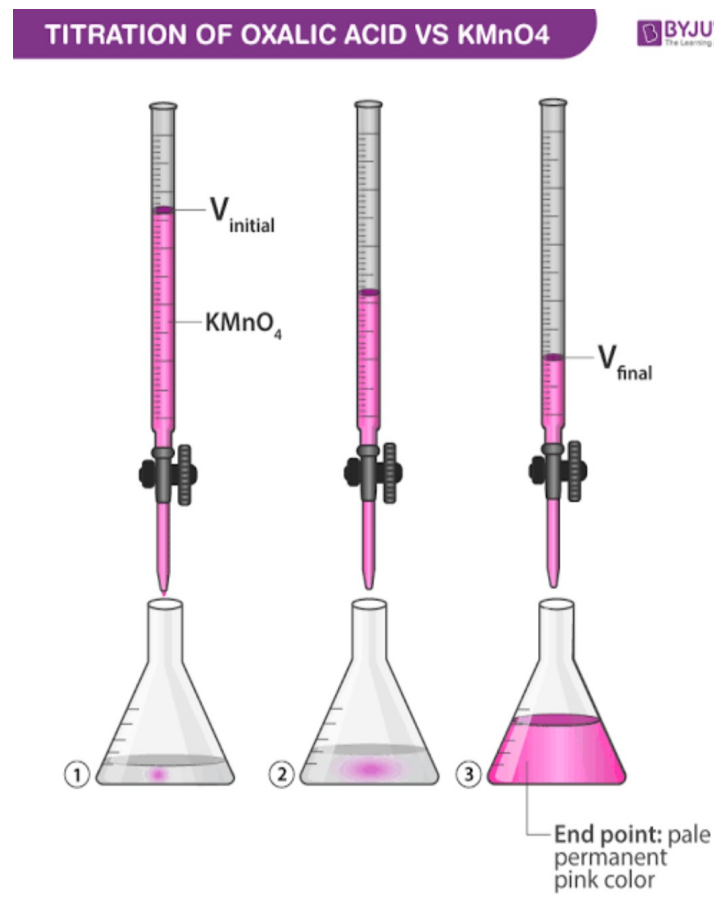
- **Diphenylamine**
- **Ferroin (1,10-Phenanthroline ferrous complex)**
- **Methylene Blue**
- **Starch-Iodine Complex**

POTASSIUM PERMANGANATE METHOD

It is a type of redox titration where **potassium permanganate** acts as the titrant. This method is often used to determine the concentration of reducing agents in a solution.

Key Points:

- **Strong Oxidizing Agent:** KMnO_4 is a strong oxidizing agent that reacts with various reducing agents.
- **Self-Indicator:** KMnO_4 has a strong purple color, which acts as a self-indicator because it turns colorless when reduced without the need for an additional indicator, indicating the end point of the titration.
- **Reaction Conditions:** The titration is typically carried out in acidic conditions (commonly using sulfuric acid, H_2SO_4) to maintain the oxidizing strength of KMnO_4 and improve its sensitivity.



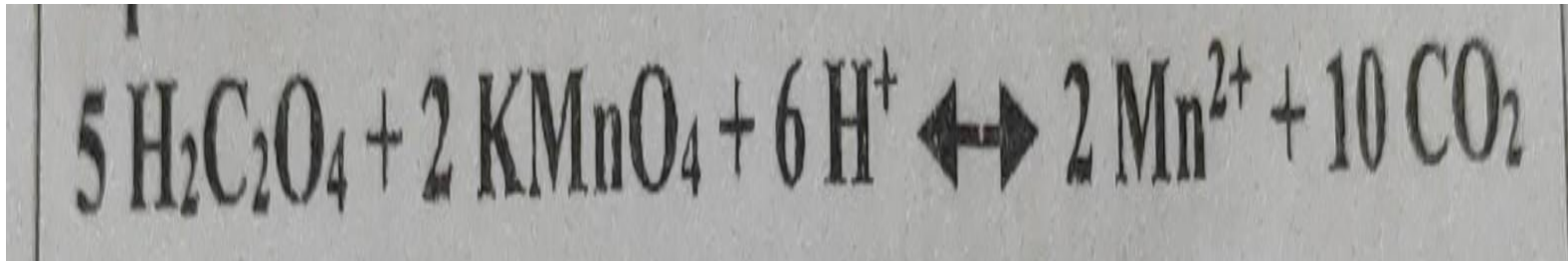
The process involves several steps:

- **Preparation:** The solution containing the reducing agent is prepared, and a few drops of sulfuric acid are added to ensure acidic conditions.
- KMnO_4 solution of known concentration is prepared as the titrant.
- **Titration:** The KMnO_4 solution is slowly added to the reducing agent solution while continuously stirring.
- As KMnO_4 is added, it reacts with the reducing agent, and the purple color of KMnO_4 fades because it is reduced to Mn^{2+} , which is colorless.
- **End Point Detection:** The titration continues until the addition of a single drop of KMnO_4 causes a persistent **pale pink color**, indicating that all the reducing agent has reacted and there is a slight excess of KMnO_4 .

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Example:

For the titration of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$) with KMnO_4 :



- KMnO_4 (purple) is reduced to Mn^{2+} (colorless).
- Oxalic acid is oxidized to carbon dioxide (CO_2).

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Advantage

- **Self-Indicating:** KMnO_4 is a self-indicator, providing a clear color change (from purple to colorless and then to faint pink) at the endpoint without the need for additional indicators.
- **Strong Oxidizing Agent:** It reacts with a variety of reducing agents, making it versatile for different types of analyses.
- **Stable Solution:** KMnO_4 solutions are relatively stable when stored properly, allowing for longer usability.
- **Wide Application Range:** It can be used in various fields, including environmental analysis, food chemistry, and pharmaceuticals.

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Disadvantage

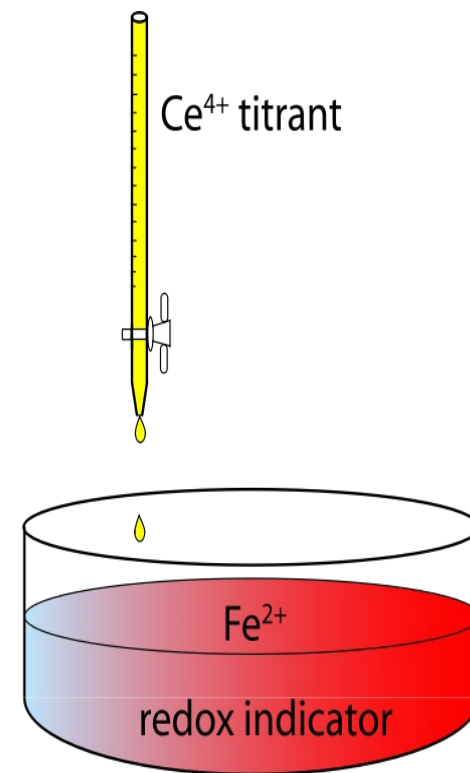
- **Sensitivity to pH:** The reaction is pH-dependent; acidic conditions are usually required for accurate results, which can complicate some analyses.
- **Oxidation of Other Species:** KMnO_4 can oxidize other substances in the solution (e.g., organic compounds), potentially leading to interference and inaccurate results.
- **Limited to Redox Reactions:** The method is applicable only for redox reactions, which may limit its use for other types of titrations (e.g., acid-base titrations).

Ceric sulfate method

- The ceric sulfate method of titration is a redox titration technique that utilizes cerium(IV) sulfate ($\text{Ce}(\text{SO}_4)_2$) as an oxidizing agent. This method is commonly used to determine the concentration of reducing agents, such as iron(II) ions, as well as other substances that can be oxidized by cerium(IV).

Key Points:

- Strong Oxidizing Agent:** cerium(IV) sulfate is a strong oxidizing agent in acidic medium that reacts with various reducing agents.
- Self-Indicator:** due to the yellow color of Ce^{4+} (oxidizing form) changing to colorless Ce^{3+} (reduced form). This property eliminates the need for external indicators, making the titration process more straightforward.
- Reaction Conditions:** The titration is typically carried out in acidic conditions (commonly using sulfuric acid, H_2SO_4) to maintain the oxidizing strength of cerium(IV) sulfate.



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The process involves several methods:

- **Standardization:** Before using ceric sulfate as a titrant, it must be standardized using a known reducing agent, such as iron(II) sulfate.
- **Preparation of the sample:** The solution containing the reducing agent is prepared, and a few drops of sulfuric acid are added to ensure acidic conditions.
- **Titration:** The cerium(IV) sulfate solution is slowly added to the reducing agent solution while continuously stirring.
- As cerium(IV) sulfate is added, it reacts with the reducing agent, and the yellow color of cerium(IV) sulfate fades because it is reduced to Ce^{3+} , which is colorless.
- **End Point Detection:** The titration continues until the addition of a single drop of cerium(IV) sulfate causes a persistent **yellow colour**, indicating that all the reducing agent has reacted and there is a slight excess of Ce^{4+} .

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Example:

For the titration of hydrogen peroxide (H_2O_2) with ceric sulfate:



- The ceric ion (Ce^{4+}) is reduced to cerous ion (Ce^{3+}).
- Hydrogen peroxide is oxidized to water (H_2O).

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Advantages

- **Stable and Reliable Titrant:** Ceric sulfate is relatively stable in acidic solutions, which makes it a reliable titrant. Its redox potential is well defined, ensuring consistent results.
- **Clear Endpoint:** The endpoint is marked by a visible color change from colorless (Ce^{3+}) to yellow (Ce^{4+}), making it easy to detect the completion of the titration without the need for an external indicator.
- **Simple Equipment and Procedure:** The titration does not require complex apparatus or procedures. Standard laboratory glassware and careful attention to titrant volume are generally sufficient.

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Disadvantages

- **Sensitivity to pH:** While ceric sulfate requires an acidic medium, the reaction may be influenced by the pH of the solution. If the pH is not maintained correctly, the results could be affected, or the titrant could become unstable.
- **Interference from Other Reducing Agents:** Other reducing agents present in the sample may interfere with the titration, leading to inaccurate results unless the sample is sufficiently purified or the interfering substances are accounted for.

IODIMETRIC METHOD

- Iodimetric methods are redox titration techniques where iodine (I_2) is used as the titrant to determine the concentration of reducing agents.
- Starch is used as an indicator in iodimetry because it forms a distinctive blue-black complex with iodine, which makes it easier to detect the end point of a titration.

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Procedure

- A standard solution of iodine is prepared
- Prepare the analyte (reducing agent)
- Add starch to the analyte as end point indicator
- Titrate the analyte with iodine solution
- **Example:**



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- **End Point Detection:** The end point of the titration is often indicated by a color change. For example, when all the reducing agent has reacted, excess iodine will react with starch indicator to form a blue complex, signaling the end point.
- Using the volume and concentration of the iodine solution, we can calculate the concentration of our reducing agent.

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Advantages

- **Sensitivity:** Iodimetric titrations can detect low concentrations of reducing agents, making them suitable for trace analysis.
- **Wide Range of Applications:** Iodimetric titrations can be applied to various substances, including organic compounds, metals, and biological samples
- **No Need for Strong Acids or Bases:** Unlike some other titrations, iodimetric titrations do not require strong acids or bases, which can simplify the procedure and reduce hazards.

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Disadvantages

- **Interference:** The presence of other reducing agents or oxidizing agents in the solution can interfere with the titration results, leading to inaccurate measurements.
- **Indicator Limitations:** While starch is commonly used as an indicator, it may not work well in all conditions or with all types of samples, leading to potential errors in endpoint determination.
- **Handling of Iodine:** Iodine can be hazardous if not handled properly, as it can cause staining and has potential health risks if inhaled or ingested.

Conclusion

- Redox titrations are versatile, reliable, and widely applicable methods in analytical chemistry, providing critical insights into the composition and quality of various substances. Each method has its specific uses and benefits, making it important to choose the right one based on the analysis requirements.

Reference

- <https://www.chemlabgenius.com/iodometry-and-iodimetry/>
- https://en.wikipedia.org/wiki/Cerium%28IV%29_sulfate?form=MG0AV3
- <https://www.intechopen.com/chapters/85563?form=MG0AV3>
- <https://unacademy.com/content/nta-ugc/study-material/pharmaceutical-analysis/acid-base-titration/?form=MG0AV3>
- <https://www.reagent.co.uk/blog/how-is-titration-used-in-the-pharmaceutical-industry/?form=MG0AV3>
- Pharmaceutical chemistry book by David watson.

Thank you!